LAPORAN

WORKSHOP SISTEM INFORMASI WEBSERVER (TIF330805)

SEMESTER 3



Tugas:
Acara 8

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PROGRAM STUDI DILUAR KAMPUS UTAMA (PSDKU)
TEKNIK INFORMATIKA KAB. NGANJUK
JURUSAN TEKNOLOGI INFORMASI
POLITEKNIK NEGERI JEMBER
TAHUN 2026

A. Judul Praktikum:

PDO, Prepared Statement, Relasi, dan CRUD Lengkap

B. Tujuan Praktikum

1. Memahami penggunaan PDO.
2. Menerapkan prepared statement.
3. Memahami relasi antar tabel.
4. Membuat CRUD data mahasiswa.
5. Membuat fitur pencarian mahasiswa.

C. Dasar Teori

1. PDO digunakan untuk menghubungkan PHP dengan database.

   Prepared Statement digunakan agar input pengguna tidak langsung menjadi query SQL sehingga lebih aman dari SQL Injection.
2. CRUD terdiri dari:

   Create → menambah data

   Read → menampilkan data

   Update → mengubah data

   Delete → menghapus data
3. JOIN digunakan untuk mengambil data dari beberapa tabel yang saling berhubungan.

D. Studi Kasus

Sistem Informasi Akademik membutuhkan fitur untuk menambah, melihat, mengubah, dan menghapus data mahasiswa.

Data disimpan di database si_akademik. Pada Acara 8 digunakan PDO dan prepared statement agar query lebih aman.

Project:

C:\xampp\htdocs\minggu4\acara8

1. Membuat Database Class/Koneksi PDO

   app/Core/Database.php

```php
<?php

class Database
{
    private static ?PDO $instance = null;
```

```
    public static function getInstance(): PDO
    {
        if (self::$instance === null) {

            $config = require __DIR__ . '/../../config/database.php';

            $dsn = "mysql:host={$config['host']};
                    dbname={$config['dbname']};
                    charset={$config['charset']}";

            self::$instance = new PDO(
                $dsn,
                $config['username'],
                $config['password'],
                [
                    PDO::ATTR_ERRMODE => PDO::ERRMODE_EXCEPTION,
                    PDO::ATTR_DEFAULT_FETCH_MODE => PDO::FETCH_ASSOC,
                    PDO::ATTR_EMULATE_PREPARES => false
                ]
            );
        }

        return self::$instance;
    }
}
```

2. Membuat MahasiswaModel

   app/Models/MahasiswaModel.php

   Model digunakan untuk proses Create, Read, Update, Delete, dan Search.

```
<?php

require_once __DIR__ . '/../Core/Database.php';

class Mahasiswa
{
    private PDO $db;

    public function __construct()
    {
        $this->db = Database::getInstance();
    }

    // READ semua mahasiswa
    public function all()
    {
```

```
        $sql = '
            SELECT
                mahasiswa.*,
                prodi.kode AS kode_prodi,
                prodi.nama AS nama_prodi
            FROM mahasiswa
            JOIN prodi ON mahasiswa.prodi_id = prodi.id
            ORDER BY mahasiswa.id DESC
        ';

        return $this->db->query($sql)->fetchAll();
    }

    // SEARCH berdasarkan nama atau NIM
    public function search($keyword)
    {
        $sql = '
        SELECT *
        FROM mahasiswa
        WHERE nama LIKE :nama
           OR nim LIKE :nim
        ORDER BY id DESC
    ';

        $stmt = $this->db->prepare($sql);

        $keyword = '%' . $keyword . '%';

        $stmt->execute([
            'nama' => $keyword,
            'nim' => $keyword,
        ]);

        return $stmt->fetchAll();
    }

    // Ambil satu data
    public function find($id)
    {
        $sql = 'SELECT * FROM mahasiswa WHERE id = :id';

        $stmt = $this->db->prepare($sql);

        $stmt->execute([
            'id' => $id,
```

```
        ]);

        return $stmt->fetch();
    }

    // Ambil data prodi
    public function getProdi()
    {
        $sql = 'SELECT * FROM prodi ORDER BY nama ASC';

        return $this->db->query($sql)->fetchAll();
    }

    // CREATE
    public function create($data)
    {
        $sql = '
            INSERT INTO mahasiswa
            (nim, nama, email, prodi_id, angkatan, status)
            VALUES
            (:nim, :nama, :email, :prodi_id, :angkatan, :status)
        ';

        $stmt = $this->db->prepare($sql);

        return $stmt->execute([
            'nim' => $data['nim'],
            'nama' => $data['nama'],
            'email' => $data['email'],
            'prodi_id' => $data['prodi_id'],
            'angkatan' => $data['angkatan'],
            'status' => $data['status'],
        ]);
    }

    // UPDATE
    public function update($id, $data)
    {
        $sql = '
            UPDATE mahasiswa
            SET
                nim = :nim,
                nama = :nama,
                email = :email,
                prodi_id = :prodi_id,
```

```
                angkatan = :angkatan,
                status = :status
            WHERE id = :id
        ';

        $stmt = $this->db->prepare($sql);

        return $stmt->execute([
            'id' => $id,
            'nim' => $data['nim'],
            'nama' => $data['nama'],
            'email' => $data['email'],
            'prodi_id' => $data['prodi_id'],
            'angkatan' => $data['angkatan'],
            'status' => $data['status'],
        ]);
    }

    // DELETE
    public function delete($id)
    {
        $sql = 'DELETE FROM mahasiswa WHERE id = :id';

        $stmt = $this->db->prepare($sql);

        return $stmt->execute([
            'id' => $id,
        ]);
    }
}
```

3. MahasiswaController

app/Controllers/MahasiswaController.php

```
<?php

require_once __DIR__ . '/../Models/Mahasiswa.php';

class MahasiswaController
{
    // READ + SEARCH
    public function index()
    {
        $model = new Mahasiswa();

        $keyword = $_GET['search'] ?? '';
```

```php
        if ($keyword !== '') {
            $mahasiswa = $model->search($keyword);
        } else {
            $mahasiswa = $model->all();
        }

        $content = __DIR__ . '/../Views/mahasiswa/index.php';

        require __DIR__ . '/../Views/layouts/main.php';
    }

    // FORM CREATE
    public function create()
    {
        $content = __DIR__ . '/../Views/mahasiswa/create.php';

        require __DIR__ . '/../Views/layouts/main.php';
    }

    // CREATE
    public function store()
    {
        $model = new Mahasiswa();

        $model->create([
            'nim' => trim($_POST['nim']),
            'nama' => trim($_POST['nama']),
            'prodi' => trim($_POST['prodi']),
            'status' => $_POST['status'],
            'dosen_id' => !empty($_POST['dosen_id'])
                ? $_POST['dosen_id']
                : null
        ]);

        header('Location: /minggu4/acara8/public/mahasiswa');
        exit;
    }

    // FORM EDIT
    public function edit($id)
    {
        $model = new Mahasiswa();

        $mahasiswa = $model->find($id);
```

```
        if (!$mahasiswa) {
            http_response_code(404);
            echo "Data mahasiswa tidak ditemukan.";
            return;
        }

        $content = __DIR__ . '/../Views/mahasiswa/edit.php';

        require __DIR__ . '/../Views/layouts/main.php';
    }

    // UPDATE
    public function update($id)
    {
        $model = new Mahasiswa();

        $model->update($id, [
            'nim' => trim($_POST['nim']),
            'nama' => trim($_POST['nama']),
            'prodi' => trim($_POST['prodi']),
            'status' => $_POST['status'],
            'dosen_id' => !empty($_POST['dosen_id'])
                ? $_POST['dosen_id']
                : null
        ]);

        header('Location: /minggu4/acara8/public/mahasiswa');
        exit;
    }

    // DELETE
    public function destroy($id)
    {
        $model = new Mahasiswa();

        $model->delete($id);

        header('Location: /minggu4/acara8/public/mahasiswa');
        exit;
    }
}
```

4. View Daftar Mahasiswa

   app/Views/mahasiswa/index.php

```php
<h2>Data Mahasiswa</h2>

<a
    href="/minggu4/acara8/public/mahasiswa/create"
    class="btn btn-primary mb-3"
>
    + Tambah Mahasiswa
</a>

<form
    method="GET"
    action="/minggu4/acara8/public/mahasiswa"
    class="mb-3"
>

    <div class="input-group">

        <input
            type="text"
            name="search"
            class="form-control"
            placeholder="Cari nama atau NIM..."
            value="<?= htmlspecialchars($_GET['search'] ?? '') ?>"
        >

        <button class="btn btn-secondary">
            Cari
        </button>

    </div>

</form>

<table class="table table-bordered">

    <thead class="table-dark">

        <tr>
            <th>No</th>
            <th>NIM</th>
            <th>Nama</th>
            <th>Prodi</th>
            <th>Status</th>
            <th>Dosen ID</th>
            <th>Aksi</th>
```

```
        </tr>

    </thead>

    <tbody>

    <?php if (empty($mahasiswa)): ?>

        <tr>
            <td colspan="7" class="text-center">
                Data tidak ditemukan.
            </td>
        </tr>

    <?php else: ?>

        <?php foreach ($mahasiswa as $no => $mhs): ?>

            <tr>

                <td><?= $no + 1 ?></td>

                <td>
                    <?= htmlspecialchars($mhs['nim']) ?>
                </td>

                <td>
                    <?= htmlspecialchars($mhs['nama']) ?>
                </td>

                <td>
                    <?= htmlspecialchars($mhs['prodi']) ?>
                </td>

                <td>
                    <?= htmlspecialchars($mhs['status']) ?>
                </td>

                <td>
                    <?= htmlspecialchars($mhs['dosen_id'] ?? '-') ?>
                </td>

                <td>

                    <a
```

```
                        href="/minggu4/acara8/public/mahasiswa/<?=
$mhs['id'] ?>/edit"
                        class="btn btn-warning btn-sm"
                    >
                        Edit
                    </a>

                    <form
                        method="POST"
                        action="/minggu4/acara8/public/mahasiswa/<?=
$mhs['id'] ?>/delete"
                        style="display:inline"
                        onsubmit="return confirm('Yakin ingin menghapus
data ini?')"
                    >

                        <button
                            type="submit"
                            class="btn btn-danger btn-sm"
                        >
                            Hapus
                        </button>

                    </form>

                </td>

            </tr>

        <?php endforeach; ?>

    <?php endif; ?>

    </tbody>

</table>
```

5. Form Tambah Mahasiswa

   app/Views/mahasiswa/create.php

```
<h2>Tambah Mahasiswa</h2>

<form method="POST" action="/minggu4/acara8/public/mahasiswa">

    <div class="mb-3">
        <label>NIM</label>
```

```
        <input
            type="text"
            name="nim"
            class="form-control"
            required
        >
    </div>

    <div class="mb-3">
        <label>Nama</label>
        <input
            type="text"
            name="nama"
            class="form-control"
            required
        >
    </div>

    <div class="mb-3">
        <label>Prodi</label>
        <input
            type="text"
            name="prodi"
            class="form-control"
            placeholder="Teknik Informatika"
            required
        >
    </div>

    <div class="mb-3">
        <label>Status</label>

        <select name="status" class="form-select">

            <option value="aktif">Aktif</option>
            <option value="cuti">Cuti</option>
            <option value="lulus">Lulus</option>

        </select>
    </div>

    <div class="mb-3">
        <label>Dosen ID</label>

        <input
```

```
            type="number"
            name="dosen_id"
            class="form-control"
            placeholder="Boleh dikosongkan"
        >
    </div>

    <button class="btn btn-primary">
        Simpan
    </button>

    <a
        href="/minggu4/acara8/public/mahasiswa"
        class="btn btn-secondary"
    >
        Kembali
    </a>

</form>
```

6. View Edit

app/Views/mahasiswa/edit.php

```
<h2>Edit Mahasiswa</h2>

<form
    method="POST"
    action="/minggu4/acara8/public/mahasiswa/<?= $mahasiswa['id'] ?>"
>

    <div class="mb-3">
        <label>NIM</label>

        <input
            type="text"
            name="nim"
            class="form-control"
            value="<?= htmlspecialchars($mahasiswa['nim']) ?>"
            required
        >
    </div>

    <div class="mb-3">
        <label>Nama</label>

        <input
```

```
            type="text"
            name="nama"
            class="form-control"
            value="<?= htmlspecialchars($mahasiswa['nama']) ?>"
            required
        >
    </div>

    <div class="mb-3">
        <label>Prodi</label>

        <input
            type="text"
            name="prodi"
            class="form-control"
            value="<?= htmlspecialchars($mahasiswa['prodi']) ?>"
            required
        >
    </div>

    <div class="mb-3">
        <label>Status</label>

        <select name="status" class="form-select">

            <option
                value="aktif"
                <?= $mahasiswa['status'] == 'aktif' ? 'selected' : ''
?>
            >
                Aktif
            </option>

            <option
                value="cuti"
                <?= $mahasiswa['status'] == 'cuti' ? 'selected' : '' ?>
            >
                Cuti
            </option>

            <option
                value="lulus"
                <?= $mahasiswa['status'] == 'lulus' ? 'selected' : ''
?>
            >
```

```
                Lulus
            </option>

        </select>
    </div>

    <div class="mb-3">
        <label>Dosen ID</label>

        <input
            type="number"
            name="dosen_id"
            class="form-control"
            value="<?= htmlspecialchars($mahasiswa['dosen_id'] ?? '')
?>"
        >
    </div>

    <button class="btn btn-success">
        Update
    </button>

    <a
        href="/minggu4/acara8/public/mahasiswa"
        class="btn btn-secondary"
    >
        Kembali
    </a>

</form>
```

7. Layout

app/Views/layouts/main.php

```
<!DOCTYPE html>
<html lang="id">

<head>
    <meta charset="UTF-8">

    <meta name="viewport" content="width=device-width, initial-
scale=1.0">

    <title>SI Akademik</title>

    <link
```

```
        href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.3/dist/css/boo
tstrap.min.css"
        rel="stylesheet"
    >
</head>

<body>

<nav class="navbar navbar-dark bg-dark">
    <div class="container">
        <a class="navbar-brand"
           href="/minggu4/acara8/public/mahasiswa">
            SI Akademik
        </a>
    </div>
</nav>

<div class="container mt-4">

    <?php require $content; ?>

</div>

</body>
</html>
```

8. Routing

   Tambahkan route berikut pada routes/web.php:

```
<?php

$routes = [
    'GET' => [
        '/' => [
            'controller' => 'HomeController',
            'action' => 'index',
        ],

        '/login' => [
            'controller' => 'AuthController',
            'action' => 'loginForm',
        ],

        '/dashboard' => [
            'controller' => 'HomeController',
            'action' => 'dashboard',
```

```
            'middleware' => ['AuthMiddleware'],
        ],

        '/mahasiswa' => [
            'controller' => 'MahasiswaController',
            'action' => 'index',
            'middleware' => ['AuthMiddleware'],
        ],

        '/mahasiswa/create' => [
            'controller' => 'MahasiswaController',
            'action' => 'create',
            'middleware' => ['AuthMiddleware'],
        ],

        '/logout' => [
            'controller' => 'AuthController',
            'action' => 'logout',
        ],
    ],

    'POST' => [
        '/login' => [
            'controller' => 'AuthController',
            'action' => 'login',
        ],

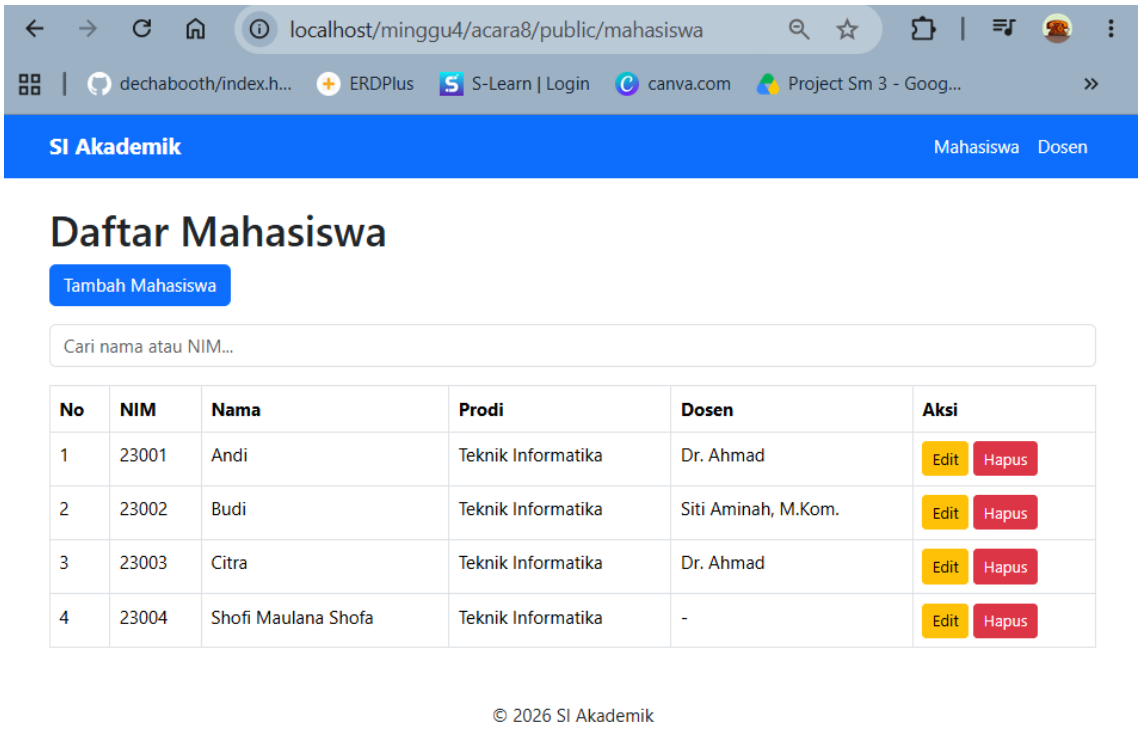
        '/mahasiswa' => [
            'controller' => 'MahasiswaController',
            'action' => 'store',
            'middleware' => ['AuthMiddleware'],
        ],
    ],
];
```

E. Hasil Studi Kasus

1. Read (Menampilkan Data)

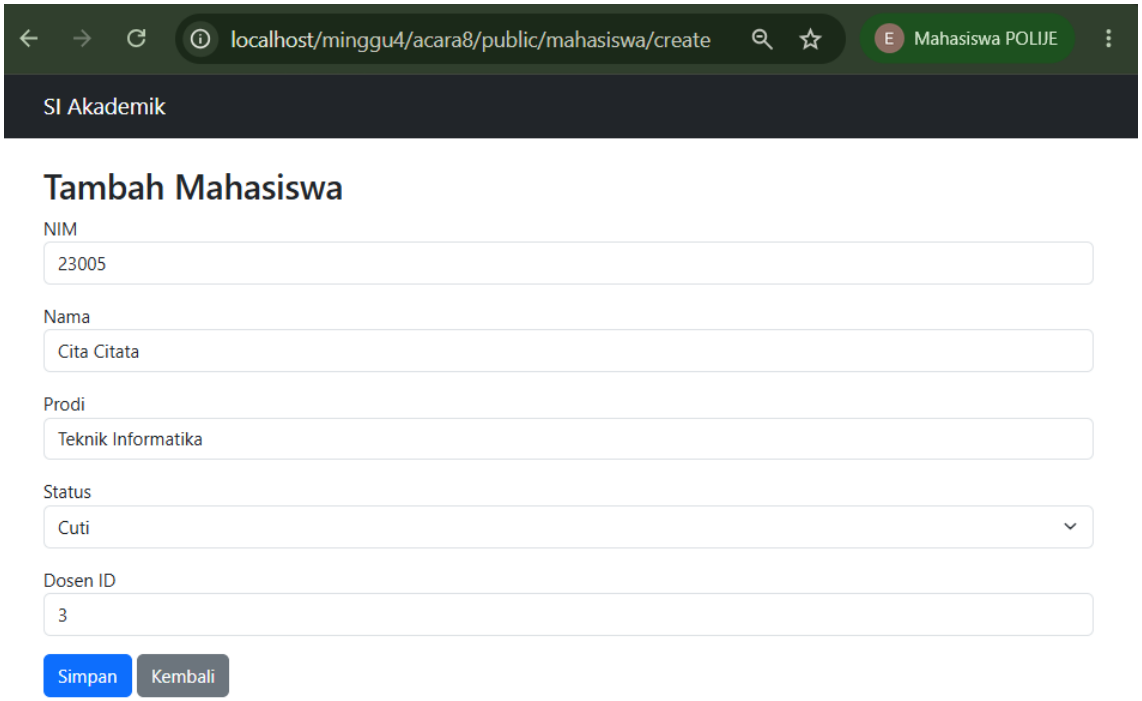
   Membuka halaman:

   Data mahasiswa berhasil ditampilkan dalam bentuk tabel.

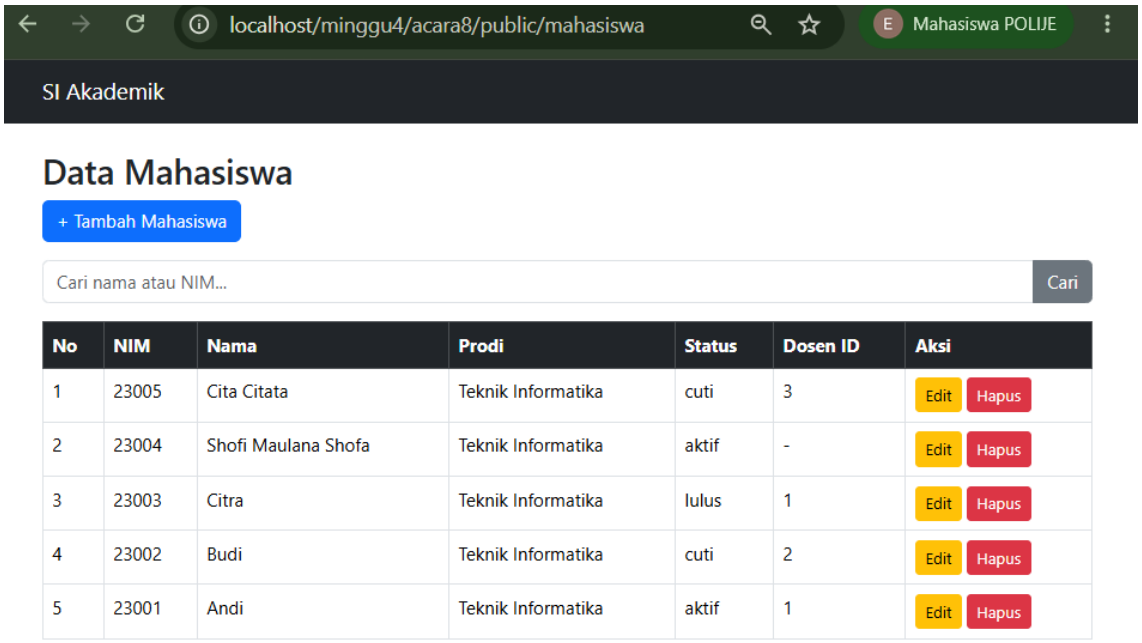


2. Create (Tambah Data)

   Menambahkan data mahasiswa:

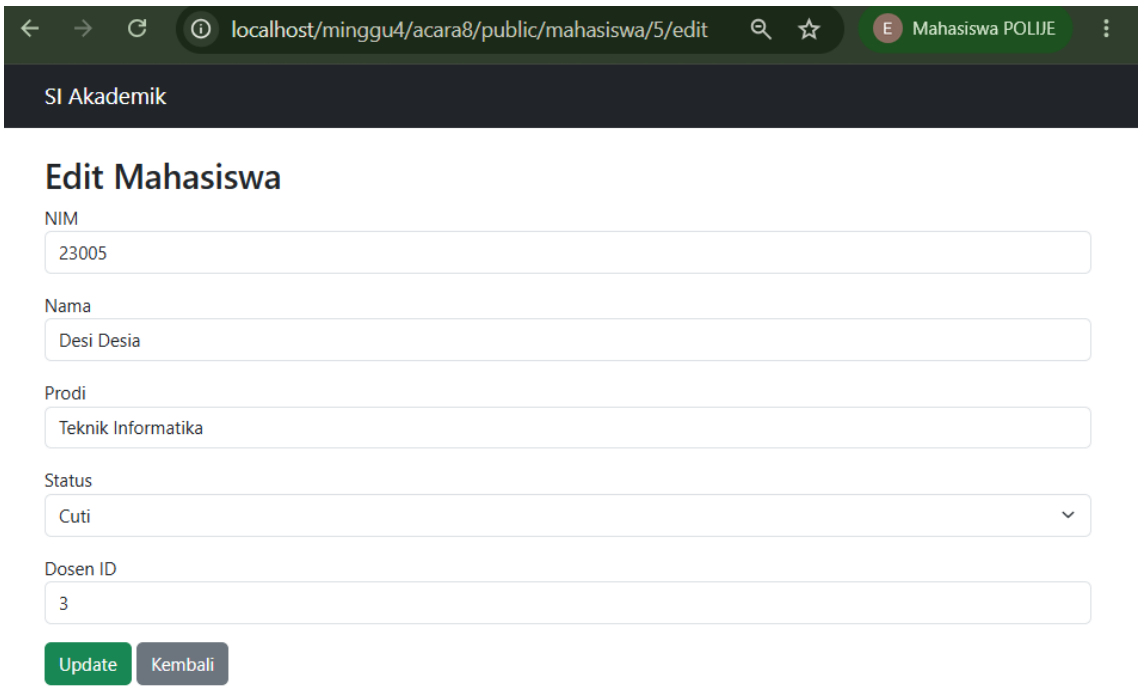


   Data berhasil disimpan dan muncul pada halaman daftar mahasiswa.

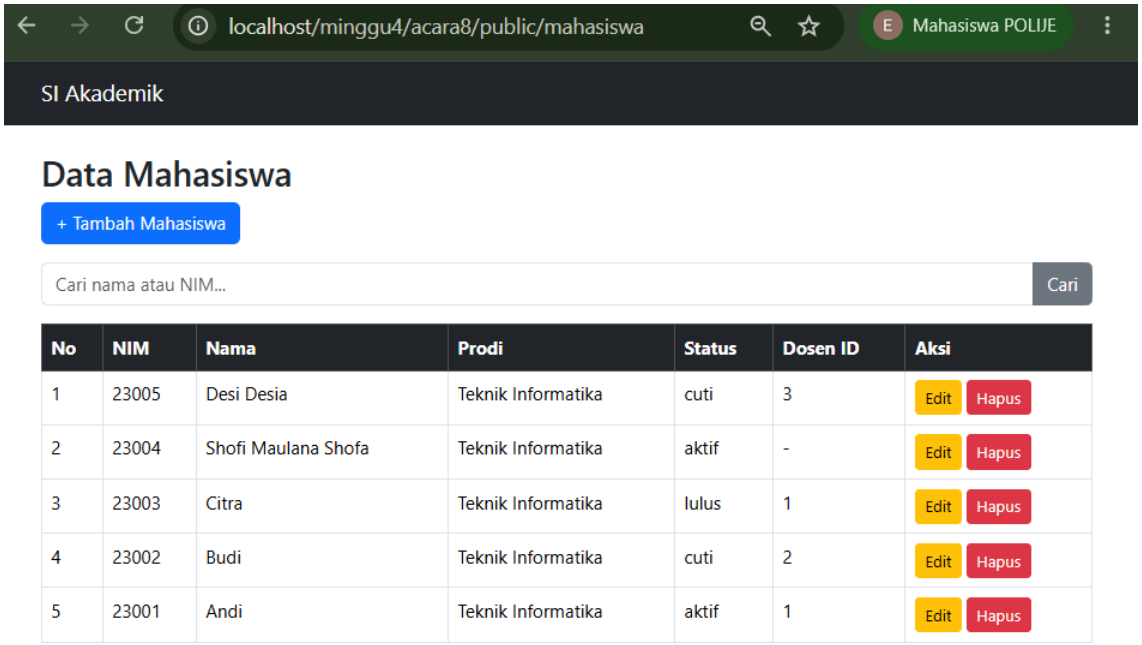


3. Update (Mengubah Data)

   Mengubah nama Cita Citata menjadi Desi Desia.



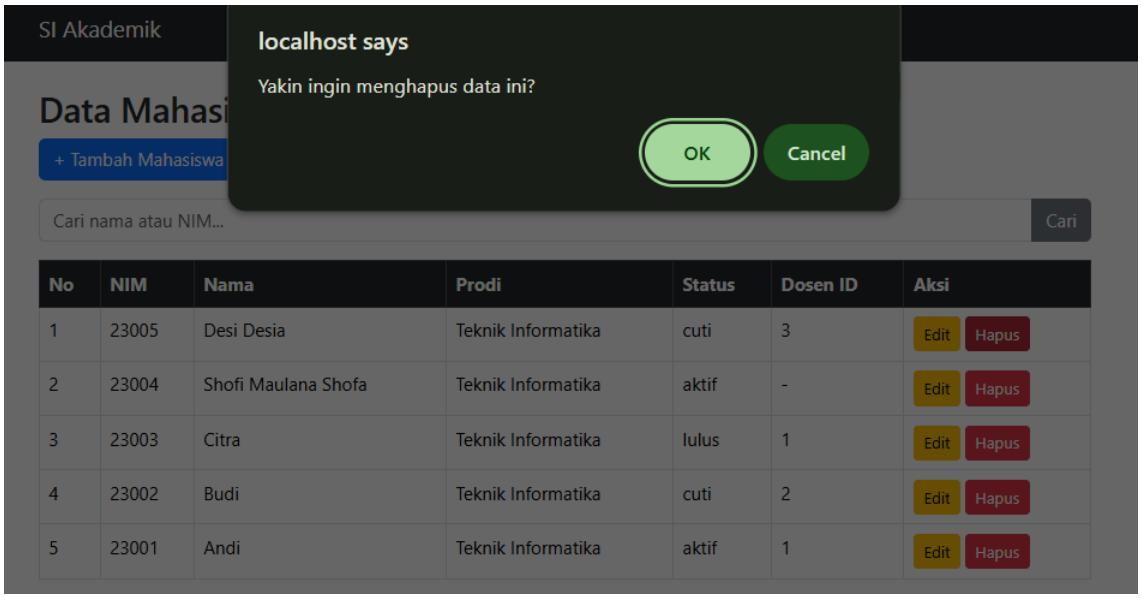
   Data berhasil diperbarui dan perubahan tampil pada tabel.



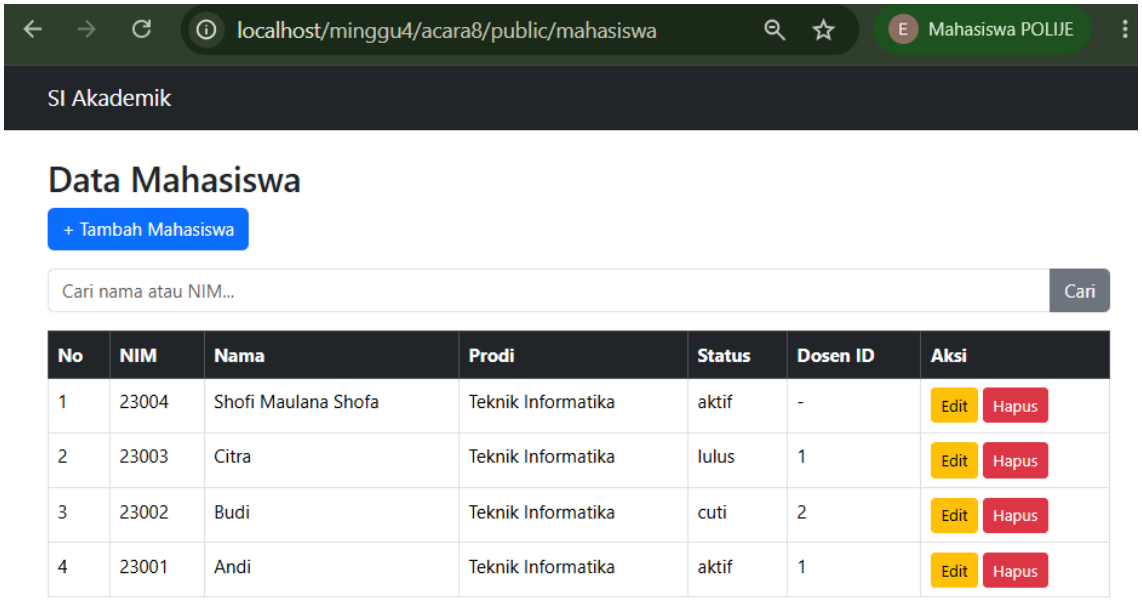
4. Delete (Menghapus Data)

   Menghapus data Desi Desia.

   Muncul konfirmasi penghapusan dan data berhasil dihapus dari tabel.



   Data berhasil dihapus,



F. Tugas Mandiri

Fitur Pencarian Mahasiswa

Pencarian dibuat berdasarkan nama atau NIM menggunakan LIKE dan prepared statement.

App/minggu4/acara8/app/Models/Mahasiswa.php

Method yang digunakan:

```
public function search($keyword)
    {
        $sql = '
        SELECT *
        FROM mahasiswa
        WHERE nama LIKE :nama
           OR nim LIKE :nim
        ORDER BY id DESC
    ';

        $stmt = $this->db->prepare($sql);

        $keyword = '%' . $keyword . '%';

        $stmt->execute([
            'nama' => $keyword,
            'nim' => $keyword,
        ]);

        return $stmt->fetchAll();
    }
```

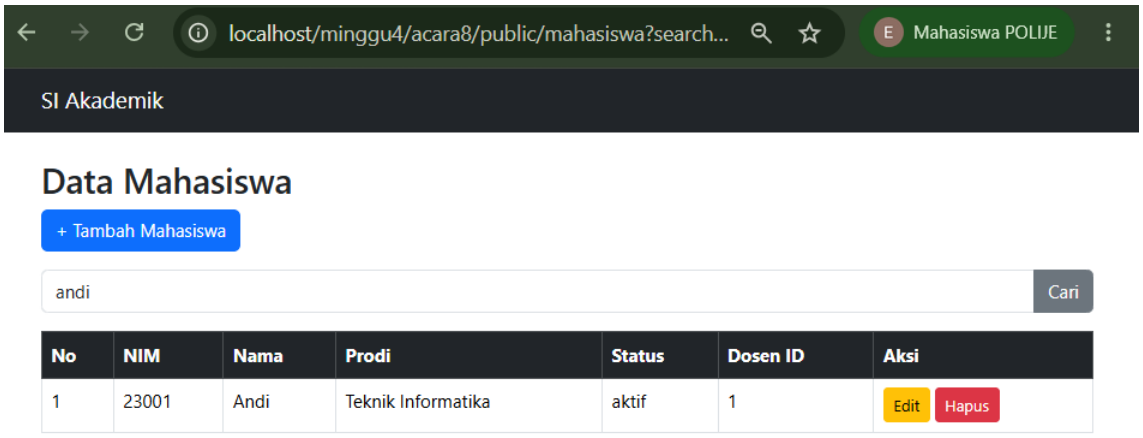
Pada halaman mahasiswa terdapat input

```
        <input
            type="text"
            name="search"
            class="form-control"
            placeholder="Cari nama atau NIM..."
            value="<?= htmlspecialchars($_GET['search'] ?? '') ?>"
        >
```

Hasil Tugas Mandiri:

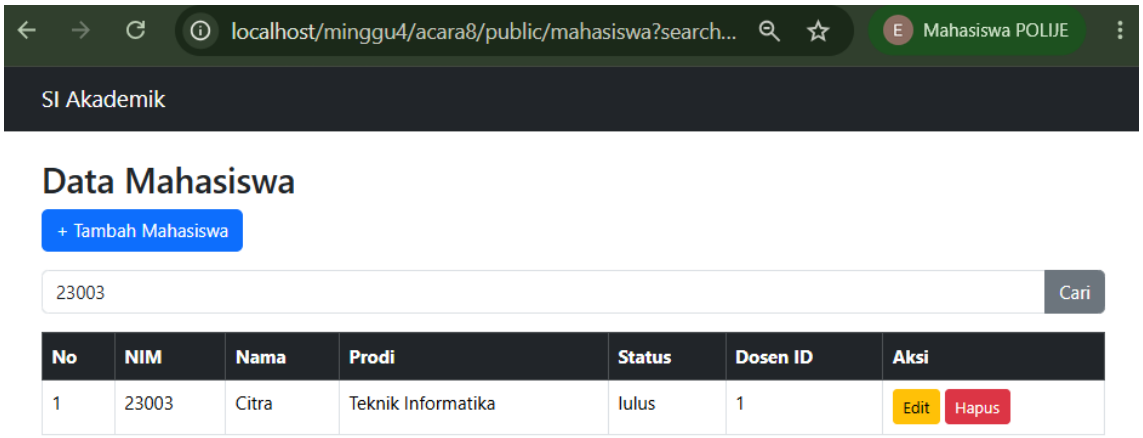
Uji Coba Search

1. Pencarian berdasarkan nama:



Hasil menampilkan data mahasiswa dengan nama yang sesuai.

2. Pencarian berdasarkan NIM:



G. Analisis

Pada Acara 8, PHP berhasil menggunakan PDO dan prepared statement untuk mengakses database. CRUD mahasiswa dapat dilakukan melalui website, sedangkan JOIN digunakan untuk menampilkan nama dosen. Fitur pencarian juga menggunakan LIKE dengan prepared statement sehingga input pengguna lebih aman.

H. Kesimpulan

PDO dan prepared statement berhasil diterapkan pada aplikasi. Data mahasiswa dapat ditambah, ditampilkan, diubah, dan dihapus melalui CRUD. Fitur pencarian berdasarkan nama dan NIM juga berhasil dibuat.